

Amortized Bayesian inference on generative dynamical network models of epilepsy using deep neural density estimators

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ABSTRACT

Whole-brain modeling of epilepsy combines personalized anatomical data with dynamical models of abnormal activities to generate spatio-temporal seizure patterns as observed in brain imaging data. Such a parametric simulator is equipped with a stochastic generative process, which itself provides the basis for inference and prediction of the local and global brain dynamics affected by disorders. However, the calculation of likelihood function at whole-brain scale is often intractable. Thus, likelihood-free algorithms are required to efficiently estimate the parameters pertaining to the hypothetical areas, ideally including the uncertainty. In this study, we introduce the simulation-based inference for the virtual epileptic patient model (SBI-VEP), enabling us to amortize the approximate posterior of the generative process from a low-dimensional representation of whole-brain epileptic patterns. The state-of-the-art deep learning algorithms for conditional density estimation are used to readily retrieve the statistical relationships between parameters and observations through a sequence of invertible transformations. We show that the SBI-VEP is able to efficiently estimate the posterior distribution of parameters linked to the extent of the epileptogenic and propagation zones from sparse intracranial electroencephalography recordings. The presented Bayesian methodology can deal with non-linear latent dynamics and parameter degeneracy, paving the way for fast and reliable inference on brain disorders from neuroimaging modalities.

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1. Introduction

Epilepsy is the second most common neurological disorder, affecting 50 million people worldwide (World Health Organization, 2020). In focal epilepsy, seizures originate in a network of hyperexcitable regions referred to as epileptogenic zone (EZ; Bartolomei et al. (2017)), and then propagate to a secondary connected network, the so-called propagation zone (PZ; Proix,

Bartolomei, Guye, and Jirsa (2017)). Drugs are used as a long-term therapeutic treatment, however, 22% to 40% of patients become resistant to medication (Beghi, 2020; Kwan et al., 2010). Resective surgery aims to remove the part of the brain responsible for seizure genesis and is used for such patients (Cardinale et al., 2019). While various methods of intracranial EEG recording exist, the method of choice for many surgical procedures is stereoelectroencephalography (SEEG; Bancaud et al. (1970)). SEEG consists of invasive implantation of intracerebral electrodes (generally 0.8 mm diameter) with contacts (2 mm long and 1.5 mm apart) targeting different brain areas. The SEEG approach, unlike other intracranial recording methods, allows simultaneous recording from multiple distributed and deeper sites (Cardinale, Casaceli, Raneri, Miller, & Russo, 2016). Critically, the outcome of surgery based on SEEG recordings, depends on the accuracy of the initial hypotheses (e.g., the spatial distribution of EZ/PZ). Reliable model-based identification of EZ/PZ from whole-brain recordings is challenging due to the large dimensionality of the parameter space, the non-trivial brain network effects, the non-linearity involved in spatio-temporal brain organization, and the uncertainties of the model components. Moreover, the dynamics of source states are unobservable, leading to non-identifiability issues for

Abbreviations: VEP, virtual epileptic patient; TVB, the virtual brain; EZ, epileptogenic zone; PZ, propagation zone; HZ, healthy zone; SEEG, stereoelectroencephalography; MCMC, Markov chain Monte Carlo; HMC, Hamiltonian Monte Carlo; VI, variational inference; ANNs, artificial neural networks; SBI, simulation-based inference; NFs, normalizing flows; MAFs, masked autoregressive flows; NSFs, neural spline flows; MDNs, mixture density networks; SNPE, sequential neural posterior estimation; SNLE, sequential neural likelihood estimation; SNRE, sequential neural ratio estimation

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parameter estimation using measurements at the sensor-level, such as SEEG recordings. Due to the spatial sparsity of electrode implantation as well as the source mixing at the sensors, there is no unique and bijective mapping between the activities at source space and the measurements at sensor space, rendering the source localization a severely ill-posed inverse problem.

Bayesian inference is a principled method for statistical inference that updates the prior belief about a hypothesis with new evidence from observed data, thus naturally characterizing uncertainty over unknown quantities. Bayesian inference from SEEG recordings takes into account the uncertainties in the forward model such as the Virtual Epileptic Patient (VEP; [Jirsa et al. \(2017\)](#)), by allowing constrained variations around model components, such as the patient-specific connectome (brain network connectivity), the lead-field matrix (mapping the source activities to sensor signals), and the epileptogenicity of brain regions (bifurcation parameter). Given a set of observations, the generative model is a low-dimensional probabilistic formulation of the mechanisms by which the observed data are generated through some hidden states and unknown parameters ([Daunizeau, Friston, & Kiebel, 2009](#); [Hashemi et al., 2020](#)). Here, the generative model of epilepsy is characterized by inferring the VEP model parameters given the SEEG recordings. However, this requires marginalizing over the unobserved latent variables, which is generally intractable, leading to challenging inverse problems. Markov chain Monte Carlo (MCMC; [Gelman, Robert, Chopin, and Rousseau \(1995\)](#)) is a non-parametric method that requires explicit evaluation of the likelihood function and is asymptotically unbiased for sampling from the posterior distribution. However, evaluation of the target distribution using MCMC can be prohibitive in high-dimensional spaces, often resulting in the rejection of many proposals. Another key disadvantage of directly using MCMC to fit SEEG data is the lack of amortization. To evaluate evidence for different clinical hypotheses in personalized medicine, the amortization strategies based on artificial neural networks (ANNs) can be immediately applied to arbitrary inference on new recordings without necessitating repeated training ([Gershman & Goodman, 2014](#)). After an upfront computational cost at the simulation and training steps to learn all the joint posterior distributions, new hypotheses can be evaluated efficiently without computational overhead for further simulations.

Simulation-based inference (SBI) aims to perform flexible and efficient Bayesian inference for complex models when standard methodologies cannot be applied, due to analytic or computational difficulties to calculate the likelihood function ([Cranmer, Brehmer, & Louppe, 2020](#)). In practice, traditional approximate Bayesian computation (ABC) methods ([Beaumont, Zhang, & Balding, 2002](#); [Sisson, Fan, & Beaumont, 2018](#)) for posterior estimation suffer from the curse of dimensionality, and their performance depends critically on the tolerance level in the accepted/rejected setting. An alternative approach is to utilize ANNs to either estimate the posterior directly, bypassing the need for MCMC ([Greenberg, Nonnenmacher, & Macke, 2019](#); [Lueckmann et al., 2017](#); [Papamakarios & Murray, 2016](#)), or to use synthesized likelihood ratios that require MCMC sampling, or train classifiers to extract information from the posterior ([Durkan, Murray, & Papamakarios, 2020](#); [Hermans, Begy, & Louppe, 2020](#)). Rather than using model simulation during MCMC sampling, ANNs can be trained as parametric function approximators to learn the (log) likelihood function from a set of existing simulations generated with random parameters drawn from a prior distribution. Normalizing flows (NFs) are a family of generative models that convert a simple initial distribution into any complex target distribution, where both sampling and density evaluation can be efficient and exact ([Kobyzev, Prince, & Brubaker, 2020](#); [Papamakarios, Nalisnick, Rezende, Mohamed, & Lakshminarayanan, 2021](#); [Rezende &](#)

[Mohamed, 2015](#)). Here, generative modeling is an unsupervised machine learning method for modeling a probability distribution given samples drawn from that distribution. In this approach, a base probability distribution (e.g., a standard normal) is transformed into a more complex distribution (potentially multi-modal) through a sequence of invertible transformations. NFs leverage ANNs to represent these complex invertible transformations under the change of variables formula by preserving total probability to obtain the full probability distribution of the target parameter. In this context, density estimators reconstruct the probability density function (i.e., the relationship between the outcomes of a random variable and their probabilities) using a set of given data points, without having access to all the possible outcomes. Traditional density estimators, such as histograms and kernel density estimators, typically perform well only for a small number of random variables ([Liu, Xu, Jiang, & Wong, 2021](#)). In contrast, NFs such as masked autoregressive flows ([Papamakarios, Pavlakou, & Murray, 2017](#)) scale better with the number of random variables at a very low computational cost, and can incorporate domain knowledge into their design ([Papamakarios et al., 2021](#)).

In this study, we aim to efficiently estimate the full posterior distribution of VEP ([Hashemi et al., 2020](#)) model parameters (a local bifurcation parameter at each brain region characterizing the regional epileptogenicity and a global coupling parameter scaling the structural connectivity of patients), from a low dimensional representation of whole-brain epileptic patterns. The main motivation of using SBI approach is to benefit from fast and parallel simulations to perform straightforward, flexible, and efficient inference and predictions with uncertainty quantification, rather than a single point estimation ([Vattikonda et al., 2021](#)), and without the need for reparameterization methods to improve the convergence issues ([Jha, Hashemi, Vattikonda, Wang, & Jirsa, 2022](#)). This Bayesian methodology only requires forward simulations from the computer programming of a parametric stochastic simulator (also referred to as the generative model), rather than model-specific analytic calculations or exact evaluations of the likelihood function ([Gonçalves et al., 2020](#)). In particular, its amortized strategy at patient level provides fitting of new observations in the order of few seconds, allowing for rapid evaluation of clinical hypotheses.

The framework we present in this study leverages the power of NFs in SBI framework to learn the approximate likelihood for efficient Bayesian inference at the whole-brain level. In particular, the proposed approach used here for parameter inference from SEEG data has several benefits, such as (i) no need for source localization or online simulation during inference, (ii) dealing with degeneracy and potential multi-modalities, (iii) training on low-dimensional data features for faster inference of full posterior, (iv) amortized inference at patient-level. We show that training deep neural density estimators on model simulations is scalable for inferring whole-brain parameters with high spatial resolution. In particular, the low-dimensional summary statistics, which retain sufficient information about the parameters of the generative process, can readily provide inference on new data after initial training. We demonstrate the power and flexibility of our approach in accurately estimating the initiation and spread of seizures at source-level as well as sparse SEEG measurements. We also investigate the underlying causes of parameter non-identifiability and discuss straightforward methods for determining when parameters of the model can be uniquely estimated from the sparse observations. Finally, we benchmark on existing state-of-the-art algorithms for conditional density estimation embedded in SBI families, to decide which algorithm to use on empirical recordings.

The presented SBI-VEP approach is now available on TVB-inverse services on the cloud platform EBRAINS (<https://ebrains>).

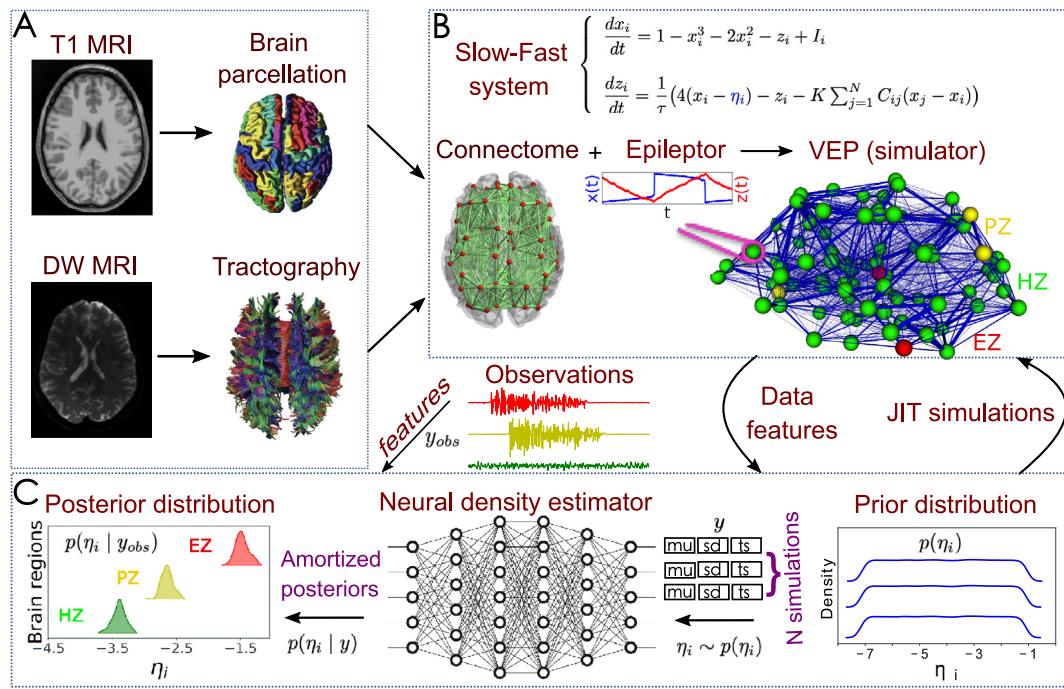


Fig. 1. The SBI-VEP workflow to estimate the amortized posterior distribution of the spatial map of epileptogenicity across different brain regions. (A) TVB reconstruction pipeline. The T1-weighted MRI images are processed to obtain brain parcellation. Diffusion-weighted (DW-MRI) images are used for tractography. With the generated fiber tracts and the regions defined by the brain parcellation, the connectome is built by counting the fibers connecting all regions. (B) The Epileptor model as a generic slow-fast dynamical system, is merged with a patient's connectome to build the VEP simulator. Using just-in-time (JIT) compilation, it allows for the fast generation of various spatio-temporal patterns. (C) SBI with deep neural density estimators. First, the model parameters are drawn randomly from a prior distribution $p(\eta_i)$. Then, the VEP simulator takes the parameters as inputs and generates a simulated dataset as an output. A class of normalizing flows such as MAF is then trained on low-dimensional data features of simulations to learn all the amortized posteriors $p(\eta_i | y)$. Finally, for new observed data y_{obs} , we can readily approximate the true posterior $p(\eta_i | y_{obs})$.

eu) to facilitate reproducible online collaboration on the Bayesian inversion of virtual brain models (Schirner et al., 2022). EBRAINS is a European digital infrastructure dedicated to neuroscience research, offering modeling and simulation services for collaborative brain research, and making data, tools, and results accessible to everyone within a protected environment (Amunts et al., 2022, 2016, 2019). The flexible probabilistic tool developed in this study may help to discover more realistic brain dynamics underlying brain diseases in a Bayesian causal inference framework.

2. Materials and methods

2.1. Individual patient data

We selected patients initially diagnosed with temporal lobe epilepsy (surgical procedure: resection, surgical outcome: seizure-free, Engel score I). The patients underwent comprehensive presurgical evaluation, including clinical history, neurological examination, neuropsychological testing, structural and diffusion MRI scanning, Stereotactic-EEG (SEEG) recordings along with video monitoring as previously described in Bartolomei, Chauvel, and Wendling (2008), Proix et al. (2017). For the selected patients, SEEG electrodes were implanted in critical regions based on the presurgical evaluation. Each electrode had 10–15 contacts (length: 2 mm, diameter: 0.8 mm, contact separation: 1.5 mm). SEEG signals were recorded with a 128-channel Deltamed system (sampling rate: 512 Hz, hardware band-pass filtering: between 0.16 and 97 Hz). To determine electrode positions, an MRI was performed after electrode implantation (T1 weighted anatomical images, MPRAGE sequence, TR = 1900 ms, TE = 2.19 ms, $1.0 \times 1.0 \times 1.0$ mm³, 208 slices) using a Siemens Magnetom Verio 3T MR-scanner. To reconstruct patient-specific connectome (DTI-MR sequence, angular gradient set of 64 directions, TR = 10.7 s, TE =

95 ms, $2.0 \times 2.0 \times 2.0$ mm³, 70 slices, b-weighting of 1000 s/mm²), diffusion MRI images were also obtained on the same scanner. The SEEG data is re-referenced using a bipolar montage, which is obtained using the difference of two neighboring contacts on one electrode. Bipolar SEEG signal is extracted from 5 s before seizure onset up to 5 s after seizure offset. The onset and offset times of the epileptic seizures are set by clinical experts. In this study, the log power of high-frequency activities is used as the target for the fitting task (Jirsa et al., 2017; Proix et al., 2017).

2.2. Network anatomy

The structural connectome was built with TVB-specific reconstruction pipeline using generally available neuroimaging software (see Fig. 1A). The current version of the pipeline evolved from a previously described version (Proix et al., 2017, 2016; Schirner, Rothmeier, Jirsa, McIntosh, & Ritter, 2015). First, the command *recon-all* from Freesurfer package (Fischl, 2012) in version v6.0.0 was used to reconstruct and parcellate the brain anatomy from T1-weighted images. Then, the T1-weighted images were coregistered with the diffusion weighted images by the linear registration tool *flirt* (Jenkinson, Bannister, Brady, & Smith, 2002) from FSL package in version 6.0 using the correlation ratio cost function with 12 degrees of freedom. The MRtrix package in version 0.3.15 was then used for the tractography. The fiber orientation distributions were estimated from DWI using spherical deconvolution (Tournier, Calamante, & Connelly, 2007) by the *dwi2fod* tool with the response function estimated by the *dwi2response* tool using the *tournier* algorithm (Tournier, Calamante, & Connelly, 2013). Next, we used the *tckgen* tool, employing the probabilistic tractography algorithm iFOD2 (Tournier, Calamante, & Connelly, 2010), to generate 15 millions fiber tracts. Finally, the connectome matrix was built by the *tck2connectome*

tool using the Desikan–Killiany parcellation (Desikan et al., 2006) generated by FreeSurfer in the previous step (the patient's brain is divided into 68 cortical regions and 16 subcortical structures, see Table S1 for label names and indices of sub-divided brain regions). The connectome was normalized so that the maximum value is equal to one (cf. Fig. S1).

2.3. VEP model

To build a whole-brain network model, aka the virtual brain model, the brain regions (network nodes) are defined using a parcellation scheme and a set of mathematical equations is placed at each network node to model the regional brain activities (Jirsa et al., 2017; Sanz-Leon, Knock, Spiegler, & Jirsa, 2015). Taking such a data-driven approach to incorporate subject-specific brain anatomy, the network edges are then represented by a structural connectivity (SC) matrix obtained from non-invasive diffusion imaging data (dMRI) of the subject. Neural mass models are flexible and physiologically realistic, providing a tractable framework for analyzing of local neural population dynamics by averaging microscopic structures and activities. Neural mass modeling has proven its efficiency in capturing the main features of brain functional behavior in a single computational framework by accounting for interactions among brain regions (Deco, Jirsa, & McIntosh, 2011; Deco, Jirsa, McIntosh, Sporns, & Kötter, 2009; Hashemi, Hutt, & Sleight, 2014, 2015). In the VEP model, the dynamics of brain regions are governed by Epileptor model (Jirsa, Stacey, Quilichini, Ivanov, & Bernard, 2014). The Epileptor is a general description of epileptic seizures that contains the complete taxonomy of system bifurcations to realistically reproduce the dynamics of onset, progression, and offset of seizure-like events (Saggio et al., 2020). The full Epileptor comprises five state variables coupling two oscillatory dynamical systems on three different timescales (Hashemi et al., 2020; Jirsa et al., 2017). By applying averaging methods, the effect of the second neuronal ensemble of Epileptor is negligible (Hashemi et al., 2021; Proix, Bartolomei, Chauvel, Bernard, & Jirsa, 2014). Then motivated by Synergetic theory (Haken, 1977; Jirsa & Haken, 1997) and under the assumption of time scale separation, the fast variables rapidly collapse on the slow manifold (Hashemi et al., 2021, 2020). This adiabatic approximation yields a 2D reduction of VEP model as follows:

$$\begin{aligned}\dot{x}_i &= 1 - x_i^3 - 2x_i^2 - z_i + I_i \\ \dot{z}_i &= \frac{1}{\tau}(4(x_i - \eta_i) - z_i - K \sum_{j=1}^N C_{ij}(x_j - x_i)),\end{aligned}\quad (1)$$

where x_i and z_i indicate the fast and slow variables corresponding to i th brain region, respectively, and the set of unknown η_i is the spatial map of epileptogenicity. The network nodes are coupled by a linear diffuse approximation of permittivity coupling through $K \sum_{j=1}^N C_{ij}(x_j - x_i)$, which includes a global scaling factor K , and the patient's connectome C_{ij} . The 2D reduction of the VEP model allows for faster inversion while enabling us to predict the envelope of fast discharges during the ictal seizure states (i.e., onset, propagation, and offset of seizure patterns). Additionally, it provides a scheme for inferring slow changes, such as extracellular ion concentrations or synaptic efficacy, in biophysical parameters that control fluctuations of fast neuronal states (e.g., seizure activity) (Jafarian, Litvak, Cagnan, Friston, & Zeidman, 2020; Jafarian, Zeidman, Wykes, Walker, & Friston, 2021).

2.4. Spatial map of epileptogenicity

In the VEP, each brain region can trigger seizures depending on its excitability value (node dynamics) and the connectivity to other regions (network coupling). In this study, depending on the excitability parameter η , the different brain regions are classified into three main types:

- **Epileptogenic zone (EZ):** if $\eta > \eta_c$, the brain region can trigger seizures autonomously and the Epileptor exhibits an unstable fixed point, allowing a seizure to happen without any dependency on network effects.
- **Propagation zone (PZ):** if $\eta_c - \Delta\eta < \eta < \eta_c$, the brain region does not trigger seizures autonomously, but it may be recruited during the seizure evolution through network effects, since its equilibrium state is close to the critical value. For these regions, a sufficiently large value of external input can cause a supercritical Andronov–Hopf bifurcation at $\eta = \eta_c$ corresponding to seizure onset; otherwise, the Epileptor is in its stable equilibrium state.
- **Healthy zone (HZ):** if $\eta < \eta_c - \Delta\eta$, the brain region stays away from triggering seizures, and all the trajectories in the phase-plane are attracted to the single stable fixed point of Epileptor.

Based on the above dynamical properties, the spatial map of epileptogenicity across different brain regions comprises the excitability values of EZ (high excitability), PZ (lower excitability values), and all other regions categorized as HZ (not epileptogenic). Note, however, that an intermediate excitability value does not guarantee that the seizures will recruit this area as part of the propagation zone, because the recruitment is also determined by various other factors including structural connectivity, network coupling, and brain state dependence. Using linear stability analysis, isolated nodes display a bifurcation at the critical value $\eta_c = -2.05$ (Olmi, Petkoski, Guye, Bartolomei, & Jirsa, 2019; Proix et al., 2014), and we set $\Delta\eta = 1.0$ (Hashemi et al., 2020).

2.5. Generative model of SEEG data

We assume a linear relation between the source activities and the measurable signals at the sensors to model the SEEG signals:

$$S_i(t) = \sum_{j=1}^{N_n} G_{ij} X_j(t), \quad (2)$$

where $S_i(t)$ is the SEEG signal at sensor $i \in \{1, 2, \dots, N_s\}$ with N_s being the total number of channels (sensors), $X_j(t)$ is the source activity in region $j \in \{1, 2, \dots, N_n\}$ with N_n being the total number of brain regions, and G_{ij} is the ij th element of the gain matrix mapping the activity at j th region to the i th electrode contact.

Here, the linear combination of source activities (fast variable x in the 2D-VEP model given by Eq. (1)) is governed by the gain matrix whose elements represent the distances of the sensors from the sources. Assuming that the generated signal decays with square of the distance from the source, the gain matrix is approximated by

$$G_{ij} = \sum_{k \in V_j} \frac{cA_k}{|\vec{x}_i^s - \vec{x}_k^v|^2}, \quad (3)$$

where V_j is the set of all vertices on the triangulate surface of region j , c is the scaling coefficient, A_k is the surface associated with vertex k , $\vec{x}_i^s$ is the position of the sensor i , and $\vec{x}_k^v$ is the

position of the vertex k . Note that the sparsity of the gain matrix (cf. Fig. S1B) creates difficulties for model inversion in terms of accurate inference of the unknown activities from different brain regions in the neighborhood of the sensor, computational time, and reliability of estimated epileptogenicity parameter (structural identifiability).

Considering the 2D reduction of VEP model given by Eq. (1) as the generative model of SEEG recordings (cf. Eq. (2)), and by fixing the initial values, the time-scale τ , and dynamical and measurement noise levels, we estimate $\theta = (\eta_1, \eta_2, \dots, \eta_{N_n}, K) \in \mathbb{R}^{N_p}$, where $N_p = N_n + 1$, with N_n as the number of brain regions.

2.6. Amortized Bayesian inference

Bayesian inference is based on the likelihood function of observations given model parameters, updated from the prior information (Bishop, 2006). The prior distribution $p(\theta)$ is typically determined before seeing the data, based on beliefs and previous knowledge about possible values of the parameters. The likelihood $p(\mathbf{y} | \theta)$ represents the probability of obtaining the data $\mathbf{y}$ given a certain set of parameter values θ (i.e., the information about the parameters available in the observed data). The likelihood function is typically intractable for high-dimensional models involving non-linear latent variables (such as the VEP model), as it corresponds to an integral over all possible trajectories through the latent space that control the generative process, i.e., $p(\mathbf{y} | \theta) = \int p(\mathbf{y}, \mathbf{x} | \theta) d\mathbf{x}$, where $p(\mathbf{y}, \mathbf{x} | \theta)$ is the joint probability density of data $\mathbf{y}$ and unmeasured latent variables $\mathbf{x}$, given parameters θ . The mapping from each observation $\mathbf{y}$ back to its representation $\mathbf{x}$ in latent space, which summarizes the high-dimensional observations, is provided by the generative model (such as the VEP). Once the form of the generative model, i.e., the joint probability distribution of the observations and the parameters $p(\mathbf{y}, \theta)$ is defined, we aim to perform inference over the parameters, for example, using a Bayesian framework.

By the product rule, the generative model can be defined in terms of the likelihood and the prior on model parameters, the product of which yields the joint density $p(\mathbf{y}, \theta) = p(\mathbf{y} | \theta)p(\theta)$. To conduct Bayesian inference, we seek the posterior distribution $p(\theta | \mathbf{y})$, which depends on both the prior and likelihood function and can be used for making predictions about future events (unseen data). Through the Bayes rule, the prior and the likelihood are combined to obtain the posterior distribution $p(\theta | \mathbf{y})$:

$$p(\theta | \mathbf{y}) = \frac{p(\mathbf{y} | \theta)p(\theta)}{p(\mathbf{y})}, \quad (4)$$

where the denominator $p(\mathbf{y}) = \int p(\mathbf{y}, \theta) d\theta = \int p(\mathbf{y} | \theta)p(\theta) d\theta$ denotes model evidence or marginal likelihood, which is a quantity of importance for model comparison. This function cannot be explicitly calculated, but can only be expressible as an analytically intractable integral. Therefore, the posterior distribution is only known up to a constant of proportionality, and since model evidence is a function of the data only, it amounts to simply a normalization term in the context of inference.

Since for most types of generative whole-brain models, the exact evaluation of likelihood function (integration over the latent variables) is often intractable (either it does not have closed-form expression, or it is computationally prohibitive to obtain), here, we use approximate inference schemes that rely on the use of deep neural networks. Our choice is motivated by the theoretical and engineering advances that make the task of training generative models significantly more approachable than in the past. More specifically, we aim to find a parametric density family $q_\phi(\theta)$ over a shared set of variational parameters ϕ , that for a given θ , best approximate the actual posterior $p(\theta | \mathbf{y})$.

Normalizing flows (NFs; Papamakarios et al. (2021), Rezende and Mohamed (2015)) are a family of methods for constructing any complex probability distribution from a simple distribution through a flow of invertible (bijective), differentiable (smooth), and parametric transformations, often implemented by ANNs. Let $\mathbf{u} \in \mathbb{R}^d$ be a random variable and $\mathbf{f} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be an invertible smooth mapping with inverse $\mathbf{f}^{-1}$. We can use $\mathbf{f}$ to transform a random variable $\mathbf{u}$ with distribution $p_u(\mathbf{u})$. By applying the change of variables formula from probability theory, the resulting random variable $\mathbf{u}' = \mathbf{f}(\mathbf{u})$ has the following probability distribution:

$$p_{u'}(\mathbf{u}') = p_u(\mathbf{u}) \left| \det \left(\frac{\partial \mathbf{f}^{-1}(\mathbf{u}')}{\partial \mathbf{u}} \right) \right|. \quad (5)$$

We can construct arbitrarily complex densities (i.e., non-Gaussian) by composing several simple maps and successively applying the transformation as given by Eq. (5). The defining property of flow-based models is that the transformation $\mathbf{f}$ must be invertible and both $\mathbf{f}$ and $\mathbf{f}^{-1}$ must be differentiable (i.e., diffeomorphisms).

NFs approximate the true posterior distribution $p(\theta | \mathbf{y})$ from a base probability distribution $p_u(\mathbf{u})$ by applying diffeomorphism transformation $\theta = \mathbf{f}(\mathbf{u})$. If an NF is able to learn the mapping between a simple prior distribution (e.g., standard normal) and a complex posterior distribution, the inverse transformation enables us to sample from the posterior by simply extracting values from the prior distribution and applying the learned transformations. By preserving the total probability and applying the change of variables formula, the posterior $p(\theta | \mathbf{y})$ is given by

$$p_\phi(\theta | \mathbf{y}) = p_u(\mathbf{f}_\phi^{-1}(\theta)) \left| \det \left(\frac{\partial \mathbf{f}_\phi^{-1}}{\partial \theta} \right) \right|, \quad (6)$$

where the first factor represents the probability density for the base distribution p_u evaluated at $\mathbf{f}_\phi^{-1}(\theta)$, and the second factor is the absolute value of the Jacobian determinant, which accounts for the change in the volume due to the transformation. Generally, the main bottleneck in using the change of variables formula is computing the determinant of the Jacobian. An important property of diffeomorphisms is that they are composable, and for such transformations, their composition is itself a diffeomorphism (Papamakarios et al., 2021). If we compose a chain of K transforms $\mathbf{f} = \mathbf{f}_1 \circ \mathbf{f}_2 \circ \dots \circ \mathbf{f}_K$, their inverse can also be decomposed in the components $\mathbf{f}^{-1} = \mathbf{f}_K^{-1} \circ \dots \circ \mathbf{f}_2^{-1} \circ \mathbf{f}_1^{-1}$ and the Jacobian determinant is the product of the determinant of each component. We can apply a sequence of diffeomorphisms $\mathbf{f}_k$ with a finite number of simple transformations $k \in 1, 2, \dots, K \in \mathbb{N}^+$ to obtain an NF:

$$\mathbf{u}_K = \mathbf{f}_K \circ \dots \circ \mathbf{f}_2(\mathbf{u}_1) \circ \mathbf{f}_1(\mathbf{u}_0), \quad \mathbf{u}_0 \sim p_0(\mathbf{u}_0), \quad (7)$$

$$p_K(\mathbf{u}_K) = p_0(\mathbf{u}_0) \prod_{k=1}^K \left| \det \left(\frac{\partial \mathbf{f}_k(\mathbf{u}_{k-1})}{\partial \mathbf{u}_{k-1}} \right) \right|^{-1}, \quad (8)$$

where $\mathbf{u}_k = \mathbf{f}_k(\mathbf{u}_{k-1})$, with initial distribution $\mathbf{u}_0$. In terms of functionality, the transformation must be flexible and expressive enough to model any desired distribution with computational efficiency; that is, calculating both forward and inverse transformations, and associated Jacobian determinants, needs to be tractable and efficient. If the transformations are conditioned on observations, the NFs can be trained to return Bayesian posterior probability estimates for any observation. Using NFs, a sample is first drawn from a base distribution, then transformed with a number of flows, and after applying K flows, the corresponding log-probability of the overall transformation is then approximated by

$$\log p_{\phi}(\theta | \mathbf{y}) \simeq \log p(\mathbf{u}_K | \mathbf{y}) = \log p_u(\mathbf{u}_0) - \sum_{k=1}^K \log \left| \det \left(\frac{\partial \mathbf{f}_k}{\partial \mathbf{u}_{k-1}} \right) \right|. \quad (9)$$

For the purpose of fitting VEP to brain epileptic patterns, we trained NFs by maximizing the log-likelihood of the observed data under our model (Eqs (1) and (2)), equivalently, minimizing the Kullback–Leibler divergence or discrepancy between the true posterior distribution $p(\theta | \mathbf{y})$ and the variational approximation of Eq. (9) denoted by a family of densities q_{ϕ} , through learning the variational parameters ϕ . In practice, this is performed by adjusting the weights ψ of a neural network F so that $q_{F(\mathbf{y}, \psi)}(\theta) \simeq p_{\phi}(\theta | \mathbf{y})$. Assuming that our dataset has N samples $\{(\theta_i, \mathbf{y}_i)\}_{i=1}^N$ with parameters drawn from the prior $p(\theta)$ and observations generated from the forward model $p(\mathbf{y} | \theta)$, the posterior $p(\theta | \mathbf{y})$ is obtained by minimizing the loss function

$$\mathcal{L}(\psi) = - \sum_{i=1}^N \log q_{F(\mathbf{y}_i, \psi)}(\theta_i) \quad (10)$$

over network parameters ψ . After the parameters of the neural networks are optimized, for new observed data $\mathbf{y}_{obs}$, we can efficiently estimate the target posterior $p(\theta | \mathbf{y}_{obs})$ by $q_{F(\mathbf{y}=\mathbf{y}_{obs}, \psi)}(\theta)$. This approach systematically outperforms other competing approaches for posterior approximation by providing a tighter variational lower bound to the marginal log-likelihood (Rezende & Mohamed, 2015). For sufficiently expressive F and q , we can construct distributions that are more complex than the base distribution, yet have easy sampling and computationally tractable evaluation.

2.7. Simulation-based inference (SBI)

Typically, simulators (i.e., a set of dynamical equations such as the VEP) implement a stochastic generative process based on a mechanistic model to produce output through a series of latent states. For many high-dimensional dynamical models that involve expensive computations such as integrals, the calculation of likelihood of the observed data given parameters can become intractable, rendering the likelihood-based inference approaches inapplicable. In this case, the simulator can be used as a black-box whose internal workings do not accessible and are not required to be differentiable, but it can generate synthetic data similar to the observed (empirical) data, allowing for inferences to be made without access to the likelihood function. Inferring the model parameters from a low-dimensional representation of synthetic data to bypass the evaluation of the likelihood function is often referred to as approximate Bayesian computation (ABC; Beaumont (2010)), likelihood-free inference (LFI; Papamakarios et al. (2021)), or simulation-based inference (SBI; Cranmer et al. (2020)). In Bayesian terminology, SBI enables us to approximate the posterior distribution of parameters of interest, conditioned on observed data, with the aid of only forward simulations, avoiding the computation of potentially intractable likelihood and its gradient. Given a prior over parameters, a stochastic simulator, and the observations, SBI returns the posterior distribution that best explains the data.

The sequential methods based on neural density estimators can be divided into three main categories: (i) sequential neural posterior estimation (SNPE; Greenberg et al. (2019), Lueckmann et al. (2017), Papamakarios and Murray (2016)), (ii) sequential neural likelihood estimation (SNLE; Lueckmann, Bassetto, Karaletos, and Macke (2019), Papamakarios, Sterratt, and Murray (2019)), and (iii) sequential neural ratio estimation (SNRE; Durkan et al. (2020), Hermans et al. (2020)). The PyTorch-based

SBI package (Tejero-Cantero et al., 2020) implements state-of-the-art algorithms for neural network-based density estimation based on sequential sampling methods (SNPE/SNLE/SNRE). The SBI package works with any simulator as long as it can be wrapped in a Python callable, and offers a flexible choice of network architectures (Gonçalves et al., 2020; Tejero-Cantero et al., 2020). The simulation step can be easily run on multiple CPU/GPU cores to benefit from effective computational parallelization, and the likelihood function is encapsulated as a feed-forward ANN, allowing for parallel evaluation by design, referred to as likelihood approximation networks (LANs; Fengler, Govindarajan, Chen, and Frank (2021)).

To perform SBI using SNPE, three inputs are needed to be specified (Gonçalves et al., 2020): (i) a prior distribution describing the possible range of parameters from which the samples can be drawn easily, (ii) a mechanistic model as a simulator that takes parameters as input and generates simulated data as output, (iii) a set of observed data (or low-dimensional data features) as the target of fitting. Taking prior distribution $p(\theta)$ over the parameters of interest θ , a limited number of simulations (N) are carried out to generate a dataset $\{(\theta_i, \mathbf{y}_i)\}_{i=1}^N$, where $\theta_i \sim p(\theta)$ and $\mathbf{y}_i$ is simulated data given model parameters θ_i . In other words, the simulated data set is a set of N independent and identically distributed samples from the generative model $p(\theta, \mathbf{y}) = p(\theta)p(\mathbf{y} | \theta)$. Then, we estimate the posterior $q_{\phi}(\theta | \mathbf{y})$ by training the NFs on the generated data set $\{(\theta_i, \mathbf{y}_i)\}_{i=1}^N$. Once the distribution is learned, for new observed data $\mathbf{y}_{obs}$, we can readily approximate the true posterior $p(\theta | \mathbf{y}_{obs})$. Active learning can be used in this approach to adaptively reduce the number of simulations over multiple simulation rounds by adding their loss terms together (Lueckmann et al., 2019). In particular, SNPE-C dynamically refines the proposals, network weights, and posterior estimates to learn how model parameters are related to the observed summary statistics of the data (Greenberg et al., 2019). Using low-dimensional sufficient statistics, this approach substantially speeds up inference, even for models where likelihood can be obtained via numerical integration, but with substantial cost. Moreover, no further simulations or training is necessary to estimate the posterior with new observations when the ANNs training is amortized.

2.8. Evaluation of posterior fit

For measuring the reliability of the inference using synthetic data, we evaluate the posterior z-scores (denoted by z) against the posterior shrinkage (denoted by s), which are defined as follows (Betancourt, 2018):

$$z = \left| \frac{\bar{\theta} - \theta^*}{\sigma_{post}} \right|, \quad (11)$$

$$s = 1 - \frac{\sigma_{post}^2}{\sigma_{prior}^2}, \quad (12)$$

where $\bar{\theta}$ and θ^* are the estimated-mean and the ground-truth, respectively, σ_{prior}^2 and σ_{post}^2 indicate the variance (uncertainty) of the prior and the posterior, respectively. The concentration of estimation towards large shrinkages indicates that all the posteriors in the inversion are well-identified, while the concentration towards small z-scores indicates that the true values are accurately encompassed in the posteriors.

2.9. Identifiability analysis

Given a mathematical model describing the observed data measured under specific conditions, different types of non-identifiability can be detected. Structural non-identifiability is

only related to the insufficient mapping information from model states to the observables, and it cannot be resolved by increasing the amount or quality of existing measurement conditions. The only remedy for this issue is qualitatively new measurements (e.g., by a change in the placement of SEEG electrodes), which alters the projection mapping (gain matrix). In contrast, practical non-identifiability can arise from a limited amount or quality of recordings under the specific experimental conditions that were used for parameter estimation (e.g., the sampling rate of SEEG recordings or the level of measurement noise). Increasing the amount and/or quality of measured data will ultimately resolve this issue in the estimation. The identifiability analysis is thus required to determine where the model does not yield reliable predictions of system dynamics.

In terms of an objective function such as chi-squared error criterion defined by

$$\chi^2(\theta) = \sum_{i=1}^{N_s} \sum_{j=1}^{N_t} \left(Y_i(t_j) - \hat{Y}_i(t_j, \theta) \right)^2. \quad (13)$$

where $Y_i(t_j)$ denotes the measured data at i th sensor with $i \in \{1, 2, \dots, N_s\}$ at time points t_j with $j \in \{1, 2, \dots, N_t\}$, $\hat{Y}_i(t_j, \theta)$ represents the corresponding model prediction, the profile likelihood (χ_{PL}^2) for each parameter individually is defined by

$$\chi_{PL}^2(\theta_i) = \min_{\theta_{j \neq i}} \left\{ \chi^2(\theta) \right\}, \quad (14)$$

to keep χ^2 as small as possible alongside the fixed θ_i , while the minimum for other parameters $\theta_{j \neq i}$ is re-obtained for each value of θ_i . The identifiability of parameter θ_i is then defined as:

- A structural non-identifiability in the parameter $\theta_i \in \theta_{sub} \subseteq \theta$ manifests as functional relation $\bar{h}$ between the parameters θ_{sub} as:

$$\chi_{PL}^2(\theta_i) = \chi_{PL}^2(\theta'_i) \Leftrightarrow \bar{h}(\theta_{sub}) = 0, \quad (15)$$

representing a manifold with constant χ^2 in parameter space ($\chi^2(\theta_i) = \text{const}$) as θ_i varies without changing the observables, due to the mapping function or the compensation by altering other parameters. Confidence intervals of a structurally non-identifiable parameter are infinite on both sides, hence, its value cannot be uniquely estimated.

- A practical non-identifiability in the parameter θ_i manifests as a likelihood-based confidence interval that is infinite in either its upper or lower bound, despite the likelihood having a unique minimum for this parameter. This indicates that the increase in $\chi^2(\theta_i)$ stays below a threshold when increasing and/or decreasing values of θ_i .
- A parameter θ_i is identifiable if

$$\chi_{PL}^2(\theta_i) = \chi_{PL}^2(\theta'_i) \Rightarrow \theta_i = \theta'_i, \quad (16)$$

indicating that a unique minimum $\chi^2(\theta_i)$ with respect to θ_i exists with finite confidence intervals in both upper and lower bounds. The profile likelihood of an identifiable parameter exceeds a threshold for both increasing and decreasing values of θ_i .

The threshold $\Delta_\alpha = \chi^2(\alpha, df)$ to determine non-identifiability is the α quantile of the χ^2 -distribution with $df = 1$ degrees of freedom, and $df = \#\theta$ being the number of parameters, corresponding to the pointwise and simultaneous confidence intervals, respectively (Raue et al., 2009; Raue, Kreutz, Maiwald, Klingmüller, & Timmer, 2011).

The identifiability analysis can also be investigated by MCMC sampling methods. From Bayesian perspective, if the posterior distribution follows the prior with no shrinkages, then, there is

no relation between the posterior samples of such parameters, and they are structurally non-identifiable (in a simple example, inference over an extraneous parameter b not included in the model $c = a$). If there is shrinkage in the posterior but with a statistical relationship (manifold) in parameter space between the sampled parameters, they are also (structurally or practically) non-identifiable (e.g., inference over parameters a and b from $c = a + b$ or $c = ab$, which manifests as a high correlation in joint posterior samples).

2.10. SBI-VEP workflow

Fig. 1 illustrates the overview of our approach, referred to as simulation-based inference for the virtual epileptic patient (SBI-VEP). This approach relies only on model-simulations to efficiently estimate the posterior distribution of the spatial map of epileptogenicity, without requiring exact likelihood evaluation. At the first step to build the SBI-VEP, the non-invasive brain imaging data such as T1-weighted MRI and Diffusion-weighted MRI (DW-MRI) are collected for a specific patient (Fig. 1A). Using the TVB-specific reconstruction pipeline, the brain of the patient is parcellated into different regions (here, using the Desikan-Killiany atlas with $N_n = 84$, comprising 68 cortical regions and 16 subcortical structures), which constitute the nodes of the brain network. The structural connectivity (SC) matrix, whose entries represent the strength of the connection between the brain regions, is derived from dMRI tractography (see Fig. S1C). This step constitutes the structural brain network component, which imposes a constraint on network dynamics (i.e., the trajectories of the latent state dynamics), allowing the hidden state dynamics to be inferred from the data. Then, the 2D reduced variant of the Epileptor neural mass model (see Eq. (1)) as a generic slow-fast dynamical system is placed at each brain network node and are coupled through SC matrix to reproduce the onset, progression, and offset of epileptic patterns across different brain regions. This combination of the brain's anatomical information (connectome) with the mathematical modeling of averaged dynamics at the level of local neural populations (e.g., 16 cm² of the cortical surface), constitutes the functional brain network model such as VEP model. Then, the VEP simulations can generate various spatio-temporal patterns of whole-brain activities as observed in brain disorders such as epilepsy, with low computational cost (Fig. 1B). Taking such a whole-brain modeling approach, the spatial map of epileptogenicity across different brain regions (EZ: epileptogenic zone, PZ: propagation zone, and HZ: healthy zone) i.e., parameters $\eta_i \in \{EZ, PZ, HZ\}$ with $i \in \{1, 2, \dots, N_n\}$ is required to be estimated from observations such as SEEG data (see Fig. S1A). In forward modeling of SEEG signals, we assume a linear relation between the non-linear latent dynamics of brain activities at the source-level (generated by Epileptor model) and the measured signals at the sensors (see Eq. (2)). Each sensor collects the source signals in its proximity and weighs them according a gain matrix that takes into account the distance from the sources (see Eq. (3)). Note that the gain (or lead-field) matrix is not of full rank in practice, as the number of sources is generally greater than the number of sensors due to the sparse placement of SEEG electrodes (see Fig. S1B). Finally, we use SBI to estimate the posterior distribution of the VEP model parameters, including the brain regional epileptogenicity η_i , which are the local bifurcation parameters, and a global coupling parameter K , which describes the scaling of the brain's structural connectivity (SC). The aim of using SBI is to obtain an efficient and accurate approximation to the true posterior of a parameterized stochastic simulator, such as the VEP. This is a dynamical system from which we can generate samples, but we cannot exactly evaluate the likelihood function.

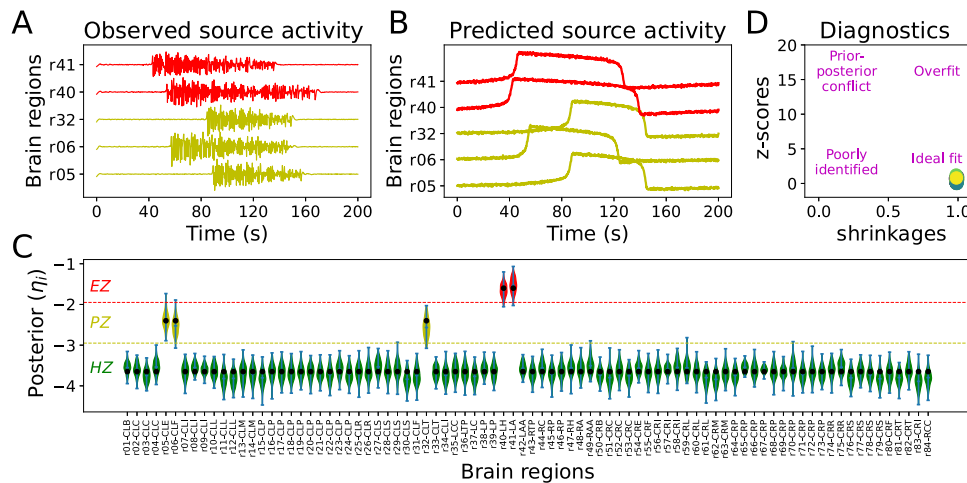


Fig. 2. Bayesian inference on the spatial map of epileptogenicity across different brain regions at source-level brain activity. (A) The observed data generated by the full 6D-VEP model with two regions as EZ (red), and three regions as PZ (yellow). (B) The predicted seizure envelope, obtained by training SNPE on a dataset containing 10^4 simulations of the reduced 2D-VEP model, while only statistical moments, peak properties, and seizure onsets were used as the data features. (C) The estimated posterior distribution of η_i for all brain regions. The filled black circles show the ground-truth used to generate the observed data. (D) The distribution of posterior z-scores versus posterior shrinkages towards the bottom right of the plot implies an ideal Bayesian inversion.

In particular, normalizing flows (NFs) embedded in the SBI approach, such as sequential neural posterior estimation (SNPE; Gonçalves et al. (2020), Lueckmann et al. (2017)), enable us to directly estimate all joint posterior distributions and bypass the need for MCMC, while also potentially capturing degeneracy or multi-modalities. To perform this step, three inputs are needed: (i) a prior distribution describing the possible range of parameters from which samples can be easily drawn, (ii) a simulator in computer code such as the VEP that takes parameters as input and generates data from drawn parameters as output, (iii) a set of low-dimensional data features for training a neural density estimator. Collecting a simulated dataset by repeatedly drawing samples from the specified prior distribution over parameters, a deep neural density estimator such as masked autoregressive flow (MAF; Papamakarios et al. (2017)) is trained to learn an invertible transformation between data features and parameters of an approximated posterior distribution. Due to this amortization strategy (i.e., a single pass through the ANNs), the SBI-VEP is able to rapidly approximate the full posterior of parameters for new observations, hence, affording clinicians the ability to evaluate different hypotheses with a negligible computational cost for the inference (in the order of a few seconds).

3. Results

3.1. The SBI-VEP against source-level epileptic patterns

We first fit the epileptic patterns at the source-level of neuronal activity to avoid any non-identifiability issue in the parameter estimation. The seizure initiation and propagation are generated using the spatial parameters with two regions in the epileptogenic zone (EZ; r40: left-hippocampus, r41: left-amygdala, with $\eta_{EZ} = -1.6$), and three regions in the propagation zone (PZ; r05: ctx-lh-entorhinal, r06: ctx-lh-fusiform, r32: ctx-lh-temporalpole, with $\eta_{PZ} = -2.4$), while all the other brain regions are set as healthy zone (HZ, i.e., not epileptogenic, with $\eta_{HZ} = -3.65$). In Fig. 2A, the activities of brain regions in EZ and PZ are shown in red and yellow, respectively. Here, we extract the envelope of signal power (high-pass filtering followed by envelope smoothing) from simulated epileptic patterns, which serves as the data feature for inference. Here we carried out 10^4 model simulations from the reduced 2D-VEP model (see

Eq. (1)) to prepare the training data set, while only the statistical moments ($m = 4$) of time-series, peak properties, and the seizure onsets were used as the data features. The parameters are drawn from a uniform prior in the ranges: $\eta_i \in \mathcal{U}(-5, -1)$ and $K \in \mathcal{U}(0, 2)$, and SNPE was run using a single round. Finally, we used a MAF (with 5 autoregressive layers, each with two hidden layers of 50 units) to learn the invertible transformation between the feature datasets generated by samples drawn from the prior distribution, and the parameters of the approximated posterior distribution. As shown in Fig. 2B, the predicted time-series are in very good agreement with the observations in terms of seizure initiation, propagation and termination (see Table S2). The estimated posterior distribution of the spatial map of epileptogenicity across different brain regions is shown in Fig. 2C. From this figure, it can be seen that the true value of η_i (shown as filled black circles) is well under the support of the estimated posterior density across different brain regions. To verify the accuracy and reliability of this result, the posterior z-scores and the posterior shrinkages are investigated from the estimation (cf. Eqs. (11) and (12)). As shown in Fig. 2D, the distribution of posterior z-scores and posterior shrinkages for all the estimated parameters substantiates the reliability of the Bayesian inference using SNPE. Notably, the estimated posteriors using SNPE are in very close agreement with those estimated using Hamiltonian Monte Carlo (HMC) algorithm by fitting the envelope of the time-series (see Fig. S2). Using SNPE, the only heavy computational task is running simulations to generate training data. However, this approach is efficient as it allows for running model simulations in parallel processors (each VEP run to simulate 100 s of whole-brain epileptic patterns takes only 0.003 s using a JIT compiler). For SBI, the model simulations can be easily run independently to create the training dataset (the more simulations, the more informative posterior), whereas HMC is amenable to embarrassingly parallel execution of only independent chains as each HMC chain itself is a simulation of Hamiltonian dynamics. Using MAF, the posterior estimation took around 1000 s and generating 200 samples from posterior took less than 1 s, whereas generating 200 samples using HMC (with an efficient non-centered form of parameterization implemented in Stan (Carpenter et al., 2017)) took around 25000 s. This indicates that for the VEP model, SNPE is 25 orders of magnitude faster than HMC. Importantly, due to amortized strategy in SBI, we can quickly estimate different

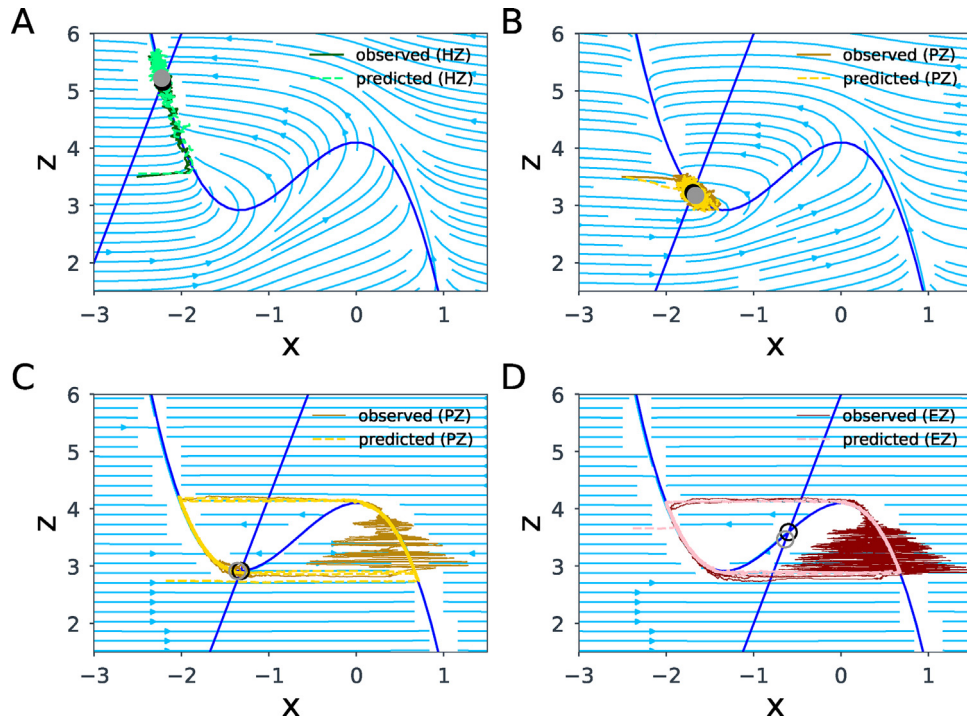


Fig. 3. The observed and predicted trajectories using 6D-VEP and 2D-VEP models, respectively, illustrated in the phase-plane for different brain regions. (A) The phase portrait of the system for healthy zone (HZ) represents a single stable fixed point, implying that this region is not epileptogenic. (B) and (C) demonstrate the phase portrait for propagation zone (PZ) with the global coupling parameter set to $K = 0$, and $K = 1$, respectively, as the candidate brain regions for seizure recruitment. Since the equilibrium in PZ is close to the bifurcation value, the spread of seizures to these regions is a complex network effect. (D) The phase portrait of the system for epileptogenic zone (EZ) demonstrates an unstable fixed point, implying that the seizure initiation is an intrinsic property of the region. The intersection of the x - and z -nullclines (colored in blue) determines the fixed point of the system. The filled and empty circles indicate the stable and unstable fixed points, respectively.

EZ/PZ hypotheses without having to carry out new simulations for the inference (by excluding the computational time for training, SNPE is 25000 times faster than HMC). See Fig. S3 for two other analyses with different spatial maps of epileptogenicity across different brain areas. An amortized inference using a MAF in SNPE with a higher measurement noise level at the source-level is shown in Fig. S4.

3.2. The SBI-VEP against phase-plane dynamics

To illustrate the capabilities of SNPE in capturing the true mechanism underlying seizure initiation and propagation from a dynamical systems theory, the phase-plane topology across different brain regions is presented in Fig. 3. In Fig. 3A, the observed and predicted brain activities at HZ are shown, where all the trajectories in the phase-plane settle into the single stable fixed point of the system (not epileptogenic). Fig. 3B and C illustrate the observed and predicted activities at PZ (candidate brain regions responsible for the seizure propagation) in the absence and with sufficiently strong network coupling, respectively. The seizure may be recruited to these regions due to the network effects, as their steady-state equilibrium is close to the bifurcation value. This indicates that seizure propagation depends on the interplay between multiple factors including the brain region's excitability (node dynamics), the individual connectome (network structure), the network coupling (parameter K), and brain state dependency (such as noise-level). For instance, in the absence of (or with weak) network coupling ($K = 0$ shown in Fig. 3B), the seizures cannot spread across the brain, whereas with sufficiently strong coupling ($K = 1$ shown in Fig. 3C) the seizure recruitment can be observed in regions defined as PZ. Lastly, Fig. 3D shows the observed and predicted brain activities at EZ, where the Epileptor model exhibits an unstable fixed point, allowing a seizure to occur

due to the brain region's intrinsic property. This implies that seizure initiation depends only on the excitability parameters of the brain regions (neither the time-scale of the system, nor the network coupling parameter). From this figure, we observed that SNPE closely follows the true trajectories in the phase-plane that controls the generative process. Notably, we can obtain similar results by training SNPE with the fixed point of the system as the data feature to estimate the spatial map of epileptogenicity in the 2D-VEP model (see Fig. S5). Moreover, we analyzed the joint posterior density distributions of all the model parameters (the bifurcation parameter η , the system time-scale τ , and initial conditions of the fast and slow variables, for different brain node types as EZ/PZ/HZ (see Fig. S6). SNPE precisely demonstrates a non-identifiability for the system time-scale in HZ, and the initial conditions in EZ and PZ, as expected from dynamical system properties. These results indicate that SNPE is able to accurately identify the stability of the equilibrium in the existence of Andronov–Hopf bifurcation from only seizure onset and therefore, explains the changing nature of seizure initiation and propagation in a non-linear system of epilepsy spread.

3.3. The SBI-VEP against SEEG data

Due to the sparse implantation of electrodes, there is no unique and bijective projection from sensors to sources, consequently, the source localization is a severely ill-posed problem. The sparsity of the gain (lead-field) matrix mapping the sources to the electrode contacts significantly impacts the precision of the resulting source reconstruction. Using the generalized (Moore–Penrose) inverse of the gain matrix G , the pseudo-inverse of the matrix $Q = G^T G$ is very sensitive to the tolerance level, i.e., the cutoff for small singular values (see Fig. S7A). To avoid such issues in source localization, we train SNPE directly on simulated

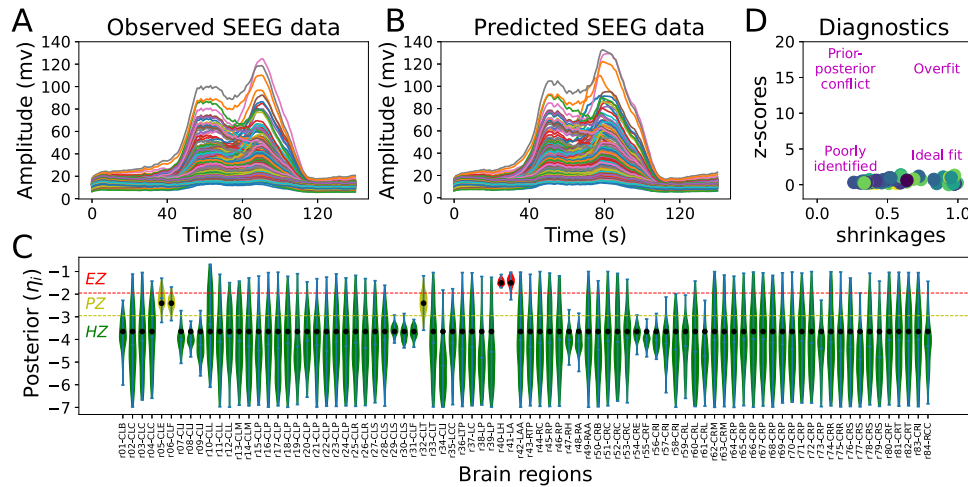


Fig. 4. Bayesian inference on the spatial map of epileptogenicity across different brain regions from the sparse SEEG signals. **(A)** The observed SEEG data with two regions in the EZ, and three regions in the PZ. **(B)** The predicted SEEG data by training SNPE on a dataset containing 10^5 simulations of the reduced 2D-VEP model, using only time-independent summary statistics as the data features. **(C)** The estimated posterior distribution of η_i for all brain regions (green: HZ, yellow: PZ, red: EZ). The filled black circles show the ground-truth used to generate the observations. **(D)** Although the regions close to the electrodes are well-identified, the zero elements in the gain matrix (due to sparse implantations) impede an ideal Bayesian inversion at whole-brain level. The poorly-identified excitability of the distant regions from electrodes results in a distribution of posterior z-scores versus posterior shrinkages with fat tails towards lower shrinkage.

SEEG data at the sensor-level to estimate the posterior of the brain parameters, assuming the lack of knowledge about seizure dynamics (onsets and offsets) in the source space.

Here, the observations are generated from the reduced 2D-VEP model (see Eq. (1)) by considering the connectome and the (low-rank) gain matrix of a patient with left focal epilepsy. A zero-mean Gaussian noise with $\sigma = 0.1$ was also added to the observations, as shown in Fig. 4A. To create the training data set, we carried out 10^5 simulations, and the parameters were drawn from a uniform prior in the ranges: $\eta_i \in \mathcal{U}(-7, -1)$ and $K \in \mathcal{U}(0, 2)$. We trained SNPE with a MAF in a single-round, while only the statistical moments ($m = 4$), peak properties, and the SEEG envelope power (the area under the curve) were used as the data features. As shown in Fig. 4B, the predicted SEEG signals are in very close agreement with the observations. From Fig. 4C, we observe that for the regions close to the electrodes (including the EZ and PZ), the posterior is tightly centered on the ground-truth parameters, whereas for many regions which are distant from the electrodes, there is no shrinkage in estimated posterior from the defined prior (see also Fig. S7B). This is due to the zero elements in the gain matrix, which conceal any impact from the activities of these regions on the observed SEEG data. The poorly-identified excitability of the distant regions from electrodes due to their non-identifiability results in the distribution of posterior z-scores versus posterior shrinkages to have fat tails towards lower shrinkage, as shown in Fig. 4D. See Figs. S8 and S9 for the detailed identifiability analyses using Frequentist and Bayesian frameworks, respectively. Despite the presence of structural and practical non-identifiability in the brain regional parameters, the posterior distribution for global coupling parameter K is accurately recovered (see Fig. S7C). The posterior predictive check (i.e., generating data from the model using the parameters drawn from the estimated posterior and then comparing them with the observed data) validates the reliability of SNPE, as it correctly fits the target SEEG envelope power (see Fig. S7D). Importantly, due to the amortized strategy, we can carry out fast Bayesian inference for arbitrary observations without the need for further simulations. See Fig. S10 for inference results on two other hypotheses about the spatial distributions of excitability parameters. An amortized inference using a MAF in SNPE for another

patient with a higher measurement noise level at the sensor-level is shown in Fig. S11.

For the sake of comparison with HMC, our non-centered form of model parameterization over state transition probabilities failed to converge to the desired distribution. HMC can deal with moderate degeneracy, but it requires extensive use of computational resources. Due to the sparsity of the gain matrix and the consequent non-identifiability issue, HMC exhibited undesirable random walk behavior, and demanded a very small value of step-size (e.g., 10^{-18}) in the leapfrog integration during Hamiltonian dynamic simulation. To efficiently fit SEEG data with HMC, a reparameterization of the model configuration space is required for convergence to the typical set.

3.4. Algorithmic benchmark on the SBI-VEP

Our algorithmic benchmark consists of existing state-of-the-art approaches for neural network-based SBI namely SNLE, SNRE, and SNPE, which target the (weighted) log-likelihood, the likelihood ratios, and directly the posteriors, respectively. The SNLE family utilizes deep neural networks to learn an approximation of the intractable likelihood, followed by MCMC sampling to obtain the posterior. SNRE approaches use classifiers to learn likelihood ratios between two choices of parameters, and MCMC is employed to obtain samples from the posterior. Rather, SNPE approaches use deep neural networks to directly target the posterior, bypassing the need for MCMC. Additionally, we use the current state-of-the-art implementations of normalizing flows (NFs) for training on simulations known as masked autoregressive flows (MAFs), neural spline flows (NSFs), and mixture density networks (MDNs). The metrics used here to rank the algorithmic performance are the accuracy in the classification of EZ/PZ/HZ across whole-brain regions, the reliability of estimation (measured by posterior z-scores versus posterior shrinkages), and the computational cost for convergence.

We first ran different flow-based algorithms (MAFs/NSFs/MDNs) embedded in SNPE to infer excitability parameters across whole-brain regions at source-level measurements, in which all the parameters are identifiable (Fig. 5A). We observed that MDN failed to converge, whereas MAF and NSF resulted in the same

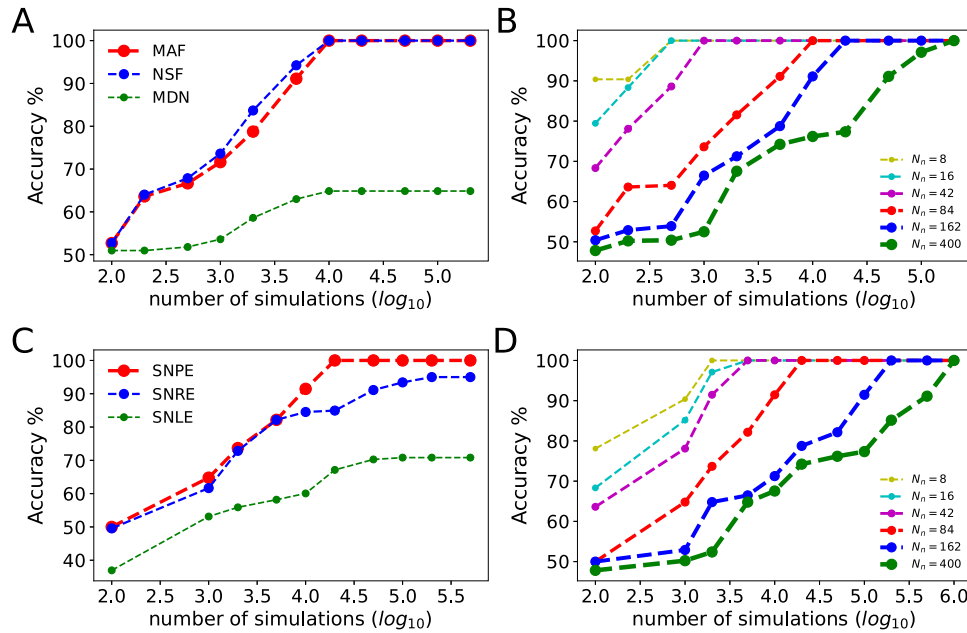


Fig. 5. The state-of-the-art algorithms in the SBI-VEP. (A) Accuracy in inference from whole-brain regions at source-level using three flow-based algorithms namely MDN, MAF, and NSF, which are embedded in SNPE. (B) Accuracy in inference from whole-brain regions at the source-level for different brain network sizes using a MAF embedded in SNPE. (C) Accuracy in estimation from sparse SEEG recording at the sensor-level using a MAF embedded in three SBI approaches known as SNLE, SNRE, and SNPE. (D) Accuracy in inference from whole-brain regions at the sensor-level for different brain network sizes using a MAF embedded in SNPE.

level of accuracy. Although NSF provided better posterior shrinkages than MAF (see Fig. S12), it is 3–5 times slower for our model. The performance of MAF in estimating the spatial map of excitability parameters with different brain network sizes is also investigated for: a toy model with $N_n = 8$ and $N_n = 16$ regions, the only left hemisphere with $N_n = 42$, whole-brain using the Desikan–Killiany parcellation with $N_n = 84$, the VEP atlas with $N_n = 162$, and the Schaefer parcellation with $N_n = 400$ regions. We observed that, with sufficient simulation budget, MAF scales to large brain network sizes when using seizure onset as the data feature (Fig. 5B). Then, we embedded a MAF in SBI families known as SNLE, SNRE, and SNPE to infer the excitability parameters from sensor-level measurements, in which some of the brain regions are non-identifiable (Fig. 5C). Our results indicate that SNPE is more accurate than other sequential algorithms in inferring the spatial map of excitability parameters, and has more posterior shrinkages (see Fig. S13). SNRE is the fastest, but it is unable to provide sufficient posterior shrinkage. SNLE showed the longest runtimes and resulted in many false positive (type I) errors (see Fig. S14). SNRE and SNLE rely on MCMC sampling to compute the posterior, which can limit their performance. With larger simulation budgets than those used for source-level data, MAF in SNPE also scales to large brain network sizes at the sensor-level measurements (Fig. 5D). These results indicate that using a MAF embedded in SNPE to directly infer the posteriors is better suited for the VEP model than alternative approaches (likelihood based and density ratios based approaches) in terms of accuracy, the level of certainty in the estimation, and computational time for convergence. See Fig. S15 for computational time involved in different steps of inference using a MAF in SNPE.

3.5. The SBI-VEP against empirical SEEG recordings

Finally, the SBI-VEP is tested against retrospective patients with drug-resistant temporal lobe epilepsy (Engel score I, seizure-free). To this end, the constructed VEP model for a patient with right temporal epilepsy is fitted against the low-dimensional data

features of empirical SEEG recordings (using a MAF in SNPE), and the inferred spatial map of epileptogenicity is compared with the clinical EZ hypothesis. The clinical classification is a binary classification that labels each region as either part of EZ or not part of EZ, whereas the VEP model prediction-based classification consists of three classes as EZ/PZ/HZ. Here, a clinical hypothesis about the EZ is incorporated in Bayesian inference as a weakly informative prior on the excitability parameter given by $\mathcal{N}(-1.5, 1.0)$ for that region, otherwise $\mathcal{N}(-3.5, 1.0)$, where $\mathcal{N}$ represents a normal distribution. Fig. 6A shows the predicted data features and the observed data features (total sensor power). A comparison of the fitted and observed SEEG envelope power is shown in Fig. S16. The inferred seizure initiation and propagation patterns for this patient are shown in Fig. 6B. Out of the six brain regions in the clinical EZ hypothesis, the model predicted three subcortical regions (r43: right-thalamus-proper, r47: right-hippocampus, r48: right-amygdala) in EZ where the seizure started, and three cortical regions (r54: ctx-rh-entorhinal, r55: ctx-rh-fusiform, r64: ctx-rh-parahippocampal) as part of PZ where only one of the regions (r55: ctx-rh-fusiform) is recruited by the seizure. Fig. 6C and D illustrate the comparison of clinical EZ hypothesis with VEP model-predicted EZ/PZ. The prediction by SBI-VEP against another drug-resistant patient is shown in Fig. 7. Using the VEP atlas ($N_n = 162$) for this patient with left temporal lobe epilepsy (Engel score I, seizure-free), out of the six brain regions in the clinical EZ hypothesis, the model predicted two cortical regions (r23: ctx-lh-parahippocampal-gyrus, r43: ctx-lh-temporal-pole) and two subcortical regions (r80: left-hippocampus, r81: left-amygdala) in EZ. Moreover, the model predicted two cortical regions (r50: ctx-lh-anterior-transverse-collateral-sulcus, r61: ctx-lh-medial-occipito-temporal-sulcus) as part of PZ where all the regions recruited by the seizure. These results indicate that the model predictions match well with the clinical hypothesis, while understanding the seizure propagation is not obvious unless accompanied by an inference on hidden source dynamics.

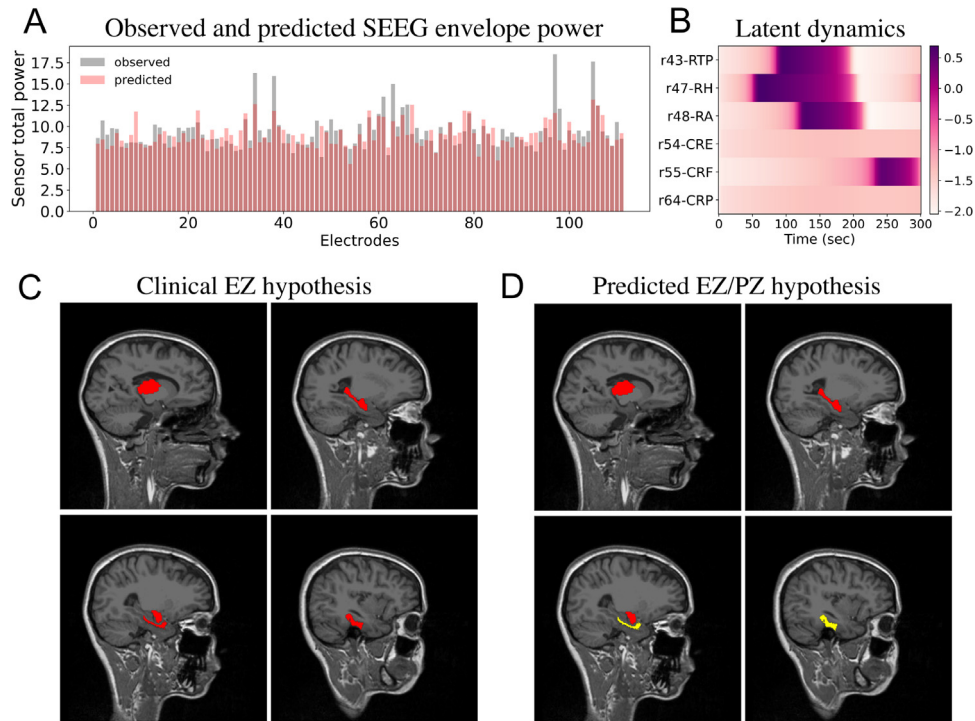


Fig. 6. Inferred seizure initiation and propagation pattern from empirical SIEG recordings of a patient with right temporal lobe epilepsy. (A) Total sensor power is used as a data feature to fit the SIEG signals. Observations are shown in gray and model predictions are shown in red (RMSE = 1.77). (B) Model-predicted seizure initiation and propagation pattern at the (unobserved) source-level brain activity. (C) Clinical EZ hypothesis. (D) Model-predicted EZ/PZ hypothesis.

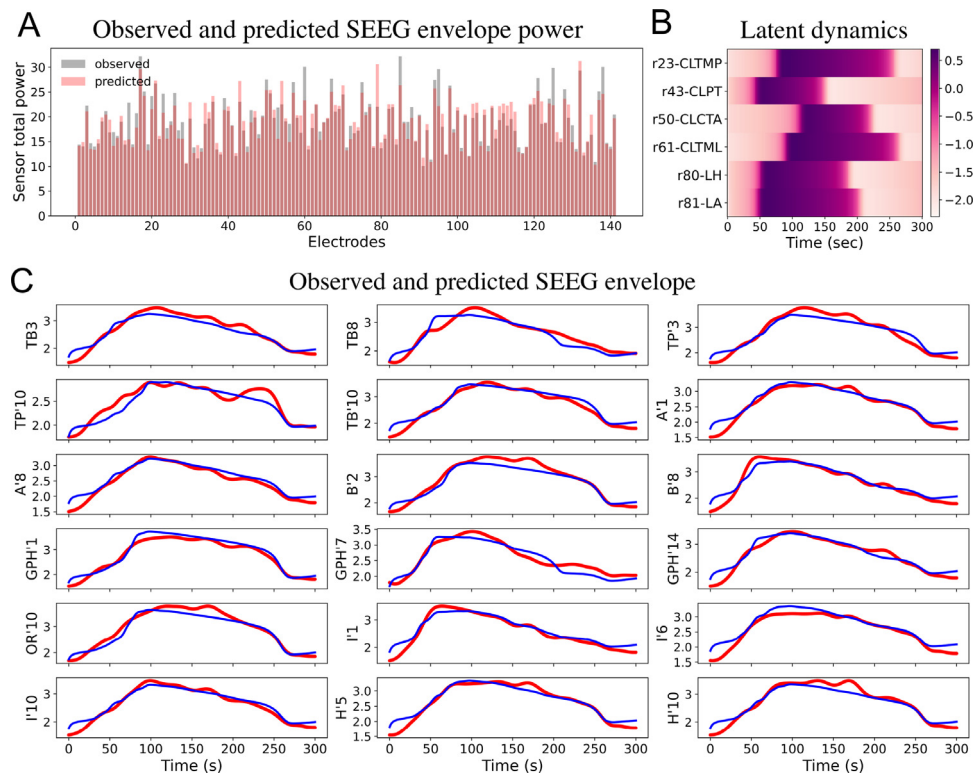


Fig. 7. Inferred seizure initiation and propagation pattern from empirical SIEG recordings of a patient with left temporal lobe epilepsy. (A) Total sensor power is used as a data feature to fit the SIEG signals. Observations are shown in gray and model predictions are shown in red (RMSE = 2.46). (B) Model-predicted seizure initiation and propagation pattern at the (unobserved) source-level brain activity. Here, r23: ctx-lh-parahippocampal-gyrus, r43: ctx-lh-temporal-pole, r50: ctx-lh-anterior-transverse-collateral-sulcus, r61: ctx-lh-medial-occipito-temporal-sulcus, r80: left-hippocampus, r81: left-amygdala. (C) Exemplary of the observed empirical (red) compared with model predicted (blue) SIEG envelope power, using MAF in SNPE.

4. Discussion

In this study, we used state-of-the-art deep neural networks for density estimation in SBI families (Greenberg et al., 2019; Lueckmann et al., 2017; Papamakarios & Murray, 2016) to amortize the posterior distribution of the spatial map of excitability by training on low-dimensional data features extracted from a limited number of VEP model simulations (see Fig. 1). We demonstrated that the SBI-VEP is able to reliably infer the distribution of EZ/PZ from whole-brain activities at the source (see Fig. 2) and sensor (see Fig. 4) levels, while we needed fewer forward simulations with having the seizure onset as the data features at source-level, compared to time-invariant summary statistics, such as SEEG log power used for sparse sensor-level data (e.g., 10 orders of magnitude for 84 brain regions). We also demonstrated that the SBI-VEP is able to accurately classify the stability of system fixed point at each brain region (EZ: unstable fixed point, PZ: close to bifurcation, and HZ: stable fixed point, as shown in Fig. 3). Note that, since we have defined seizure propagation as a brain network effect, an accurate estimation of the global coupling parameter is critically necessary to avoid misclassification of PZ as EZ (by underestimation) or HZ (by overestimation). EZ is the network of brain regions where seizure originates, and PZ is the network of brain regions that are later recruited by the seizure due to the coupling with regions in EZ. In agreement with clinical expertise, the time of recruitment was previously used to classify EZ/PZ (Proix et al., 2014, 2017; Vattikonda et al., 2021). Therefore, the PZ identification depended on a tolerance value thresholding the inferred activities at unobserved source level. In contrast, this study relied on the estimated posterior distribution, and the EZ/PZ classification was based on the bifurcation value, independent of the recruitment (see Figs. 6 and 7). Compared to the thresholding approach, this can lead to an overestimation of PZ in regions that were not recruited by the seizure due to weak coupling, or an underestimation of EZ in regions with strong coupling to the other regions.

It is known that approximate Bayesian computation (ABC) methods suffer from the curse of dimensionality and the choice of performance metric is critical (Cranmer et al., 2020). It has been shown that neural network-based approaches outperform ABC-related methods (Gonçalves et al., 2020). However, the performance ranking of these algorithms is task-dependent, and there is no uniformly best algorithm across different tasks (Lueckmann, Boelts, Greenberg, Gonçalves, & Macke, 2021). Therefore, we performed a benchmark on expressive deep neural density estimators used in different approaches to SBI that do not require explicit evaluation of likelihoods. Among the current state-of-the-art algorithms for deep neural density estimator (MAFs/NSFs/MDNs) and SBI families (SNLE, SNRE, SNPE), we found that using MAFs in a single round of SNPE is more accurate and efficient than alternatives in terms of overall accuracy, reliability of estimation, and computational cost (see Fig. 5, and Figs. S12–S15). Notably, using MAFs in SNPE scales to high-resolution brain parcellation (e.g., $N_n = 400$ brain regions shown in Fig. 5), enables us to maximize the model's predictive power in identifying the EZ/PZ, thus better supporting surgery as a viable treatment option for drug-resistant epilepsy.

The SBI-VEP approach shares a key aspect with the well-established framework for inference in neuroimaging modalities known as dynamical causal modeling (DCM; Friston, Harrison, and Penny (2003)): Bayesian inference over the parameters of a dynamical system model. However, there are key differences in practice. From the modeling perspective, the SBI-VEP is a connectome-based approach that considers structural connectivity (SC) as known parameters, which are obtained from non-invasive imaging data of individuals. Whereas DCM is furnished

with effective connectivity to explain the effects on observations induced by actual changes in interactions among brain regions (i.e., a cause–effect relationship) (Frässle et al., 2018, 2017; Gilson et al., 2020; Prando et al., 2020), which can also better fit the model to empirical data. For instance, in the SBI-VEP, seizure recruitment to a secondary subnetwork called PZ requires a strong connection with regions responsible for seizure initiation, the so-called EZ. Considering a data-driven approach, such as tractography may limit the SBI-VEP capability to fit propagation to the PZ that is weakly connected to the EZ. In contrast, using effective connectivity, there are many more parameters to fit (e.g., $N(N - 1)/2$ symmetric connections for N regions), consequently, we will be able to capture the necessary connections for recruitment. However, inference on such a high-dimensional space can easily suffer from the non-identifiability issue, unless the changes in connectivity are constrained by the prior belief that there are transitions among a small number of brain connectivity states (Zarghami & Friston, 2020). The SBI-VEP considers non-linear brain dynamics at each brain region, whereas in DCM, the system dynamics is often linearized around a fixed point on the state space equations of neuronal dynamics, and the observation equation for the measured signal (Cooray, Sengupta, Douglas, & Friston, 2016; Papadopolou et al., 2015). From the inference perspective, the SBI-VEP is equipped with state-of-the-art deep learning algorithms for statistical inference (i.e., estimating parameters of interest from observations given a model of their statistical relationship) and density estimation (i.e., estimating the joint probability density of a collection of random variables). The use of existing state-of-the-art algorithms in NFs such as MAFs allows scaling the inference to whole-brain dynamics with high parcellation resolution (e.g., $N_n = 400$ shown in Fig. 5). Notably, NFs used in the SBI-VEP approach systematically place a tighter bound on model evidence to deal with potential multimodalities and degeneracies among parameters, whereas DCM is a variational-based approach and its mean-field variant ignores –by definition– the correlation between parameters.

Taking advantage of recent advances in probabilistic programming languages for automatic Bayesian inference, such as Stan (a state-of-the-art platform for statistical modeling), the feasibility of inverting the coupled Epileptor model that best explains a patient data at the source-level activity has been shown previously (Hashemi et al., 2021, 2020). In other studies, efficient and robust inversion of seizure propagation on intracranial recordings was achieved by either simplifying the seizure dynamics using a threshold model (Sip et al., 2021), which considerably restricts the range of model dynamics, or by embedding Epileptor equations as priors on brain source dynamics and using maximum a posteriori (MAP) techniques, which does not capture the uncertainty in parameters (Vattikonda et al., 2021). These issues could possibly be addressed by sampling the whole posterior of Epileptor parameters using gradient-based and self-tuning MCMC sampling algorithms, such as the No U-Turn sampler (NUTS; Hoffman and Gelman (2014)). However, MCMC sampling of the posterior density from sparse observations such as SEEG recordings becomes computationally prohibitive, particularly when parameters in a high-dimensional model such as VEP show strong non-linear correlations or if the posterior exhibits pathological geometries such as Neal's funnel with varying curvature (Betancourt, Byrne, Livingstone, & Girolami, 2017; Vattikonda et al., 2021). Another disadvantage of directly using the MCMC is the lack of amortization. Recently, a hybrid of NFs (such as inverse autoregressive flows parameterized by neural networks) and HMC called neural-transport (NeuTra) HMC (Hoffman et al., 2019) has been proposed to correct this sort of unfavorable geometry, while it works in the amortized setting. The amortization strategy in ANNs can be applied immediately to new data without needing to be retrained.

This allows us to quickly evaluate different clinical hypotheses for a personalized model such as VEP, without needing to use additional computing power for further simulations. NFs have also been used as (nonlocal) transition operators in MCMC algorithms to substantially accelerate sampling in high-dimensional settings with unimodal and multiple relevant metastable basins (Gabri , Rotskoff, & Vanden-Eijnden, 2022). The use of hybrid techniques, such as adaptive MCMC augmented with NFs or neural variational inference with a gradient-based MCMC sampler, to benefit from the potential advantages of both approaches remains to be investigated and validated in future work.

The deep learning algorithms have been successfully used in medical image analysis (Ker, Wang, Rao, & Lim, 2017; Litjens et al., 2017), such as segmentation (Shu, Yang, Liu, Chang, & Wu, 2023), and target tracking tasks (Yuan, Chang, Li, & He, 2022; Yuan, Shu, Liu, & He, 2022), advancing clinical practice (Rasheed et al., 2020; Truong et al., 2018). Recently, Sip et al. (2023) have introduced a method using variational autoencoders (VAEs) for nonlinear dynamical system identification to infer both the neural mass model and the region- and subject-specific parameters from the functional data while respecting the known network structure. This study focused on flow-based generative models (e.g., NFs) for efficient density estimation and probabilistic inference, in the context of large-scale network modeling of brain dynamics. Generative adversarial networks (GANs) have shown tremendous success in transforming from the latent space to the data space by learning a manifold induced from a base density (Liu et al., 2021). A benchmarking study needs to be conducted to explore the power of deep generative models, including NFs, VAEs, and GANs, within the framework of large-scale brain network modeling.

In this study, we have shown that the SBI-VEP is able to efficiently estimate the joint posterior distribution of local bifurcation parameters (regional dynamics) with the global scaling parameter (network dynamics), from low dimensional data features of whole-brain activities, at source and sensor levels. However, this approach has its limitations. Here, we used the 2D reduced variant of the VEP model for inference. This reduction is limited to demonstrating only Andronov–Hopf and saddle–node bifurcations out of six types of bifurcations responsible for oscillation onset and offset (Jirsa et al., 2014; Saggio et al., 2020), which is a limitation of the model itself in demonstrating a complete taxonomy of epileptic seizures. Across patients with drug-resistant focal epilepsy, seven distinct morphological patterns have been identified at seizure-onset: low-voltage fast activity, low-frequency high-amplitude periodic spikes, sharp activity at lower 13 Hz, spike-and-wave activity, burst of polyspikes, burst suppression, and delta brush (Ferrari-Marinho, Perucca, Dubeau, & Gotman, 2016; Perucca, Dubeau, & Gotman, 2014). By high-pass filtering the sharp transients such as spikes, the 2D-VEP is then limited to detecting other oscillatory patterns associated with seizure onsets. Although removal of areas generating high-frequency oscillations has been found to be correlated with improved surgical outcome (Haegelen et al., 2013; Jacobs et al., 2010; Zijlmans et al., 2012), seizures with different morphological patterns likely have different mechanisms of generation (Ferrari-Marinho et al., 2016; Salami, L vesque, Gotman, & Avoli, 2015). The improved understanding of the mechanisms of transition from normal brain activity to seizure has the potential to improve seizure prediction (Kuhlmann, Lehnertz, Richardson, Schelter, & Zaveri, 2018). These studies emphasize that seizure patterns should be distinguished and specified before clinical practice. Nevertheless, combining high-frequency oscillations, spikes, and other biomarkers can be a better strategy than using only one marker (Cai et al., 2021; Roehri et al., 2018).

Seizures can be categorized as focal (partial) or generalized (Devinsky et al., 2018; Stafstrom & Carmant, 2015). Focal seizures involve a small portion or a focal area of the brain (simple partial), or seizures that travel from one focal area to another (complex partial), whereas generalized seizures involve both hemispheres of the brain. This study focused on predicting the initiation and spatial propagation of focal seizures in the human brain using personalized anatomical information. Although the VEP mass model is able to produce generalized seizures at source level, it is limited to accurately predict the EZ/PZ if they are located far away from the implemented sensors (e.g., in the other hemisphere). This is due to the assumption that the generated signal decays with the square of the distance from the source, thus, the gain matrix becomes zero for regions located far from the electrodes. This conceals any impact from the activities of these regions on the observed SEEG data, and no improvement in the density estimators can solve this issue. Since structural non-identifiability is independent of the accuracy of experimental data, this issue cannot be resolved by increasing the amount and/or quality of existing measurements (see Figs. S8 and S9). The remedy for this issue is to design a new setup for the measurements, such as a new SEEG implantation (e.g., using virtual surgeries; Wang et al., 2023) which alters the mapping function through the gain matrix, or reparameterizing over the model configuration space (Jha et al., 2022). In contrast, practical non-identifiability (two regions close to electrodes with the same distance) arises due to the limited amount and/or quality of observations. The remedy for this issue is to add sufficient data to the existing observables so that it provides sufficient constraining power in the latent space dynamics, yielding a unique estimation with finite confidence intervals. Importantly, SBI is able to deal with such structurally and practically non-identifiable brain regions without impeding the inference on the identifiable regions (see Fig. 4). This is a key advantage of SBI compared to gradient-based MCMC, in which convergence issues such as slow mixing and divergence can appear in the presence of degeneracy (Betancourt, 2014; Betancourt et al., 2017), making the whole sampling process inefficient. However, such algorithmic issues can be used as a powerful diagnostic tool to detect degeneracies (Betancourt, 2016a, 2016b), whereas using SBI, it is additionally necessary to perform an identifiability analysis if there was no proper shrinkage in the posterior. In such cases, reparameterization methods are necessary to facilitate efficient exploration of the posterior distribution in terms of computational time and convergence diagnostics (Jha et al., 2022). Nevertheless, finding a reparameterization that reorganizes model configurations into spaces that are more consistent with the observed data to resolve the degeneracy issue is a key challenge in statistical analysis, and it is beyond the scope of this study. It is important to note that using higher resolution for reconstructed brain regions can intensify the non-identifiability issue, thus wasting computational time and lacking information for uniquely estimating the parameter. A remedy to solve this issue is to use high resolution neural fields representing neural activity in continuous space (Proix, Jirsa, Bartolomei, Guye, & Truccolo, 2018), rather than discrete point representation of brain regions using neural mass models. An important advantage of neural field models is that they provide a more realistic source-to-sensor mapping by considering both the orientation and distance between the dipoles and the sensors (Wang et al., 2023). Moreover, neural fields consider both local connections between neighboring points on the field and global connections along white matter fibers. However, the computational cost for model inversion on high resolution neural fields with more than 260000 vertices can be numerically unstable or restrictive.

From the inference perspective, SBI is more efficient than MCMC when the calculation of likelihood is computationally prohibitive, and only a few parameters from big data need to be estimated. Generally speaking, gradient-based MCMC methods scale better with respect to the parameter dimension (Robert, Elvira, Tawn, & Wu, 2018), whereas SBI methods scale better to high-dimensional observations, as they require a low-dimensional representation of the data (Boelts, Lueckmann, Gao, & Macke, 2022; Cranmer et al., 2020). On the one hand, the hyperparameter and noise estimation can be challenging for SBI. In this study, we fixed the values of the dynamical and measurement noise levels, and a filter was applied in the signal processing step to smooth out the observations. The calculation of data features such as seizure onsets or peak assignment from data with low signal-to-noise ratios can become inaccurate, where MCMC methods can more easily address such issues (Hashemi et al., 2020). Also, the user needs to choose the type of the data summary features and the number of simulations (Gonçalves et al., 2020). Note that, the sequential methods based on neural density estimators require the loss function to be adjusted across rounds (Greenberg et al., 2019), which can be unstable, thus problematic if the proposal distribution is very different from the prior, leading to probability mass leaking into regions without prior support (Glöckler, Deistler, & Macke, 2022). In this study, we trained the sequential neural algorithms with only one round, as we observed they struggle on this problem due to leakage. The sequential inference with truncated proposals can be used to sidestep the instabilities, by drawing simulations from the prior, but rejecting them before simulation if they lie outside of the support of the approximate posterior (Deistler, Gonçalves, & Macke, 2022). Moreover, with a feasible number of simulations, the uncertainty of the estimated posterior using SBI is higher than that of the MCMC, and a wider prior range demands a larger number of simulations. However, using high performance computing, the model simulations can be easily run independently to create a large training dataset for SBI (the more simulations, the more informative the posterior), whereas MCMC is amenable to embarrassingly parallel execution of only independent chains (Hashemi et al., 2021). On the other hand, in the presence of metastability in the state-space representation, MCMC methods require either more computational cost or intricately designed sampling strategies to accurately detect state transitions in the latent space (Gabrié et al., 2022; Jha et al., 2022). Whereas SBI allows efficient Bayesian estimation in a black-box manner without access to the knowledge on the state-space representation of a system, or in a gray-box setup, which uses additional information about or from the simulator (Cranmer et al., 2020). This allows us to investigate different mechanisms of seizure generation, such as multistability, to improve the model's predictive power in future work.

In brief, providing fast simulations, the methodology used in this study can be applied to other whole-brain models, since it requires neither model nor data features to be differentiable. However, finding low-dimensional yet sufficiently informative data features that can address parameter degeneracies is the key challenge. The applications of SBI to estimate brain dynamics considering the existence of multiple fixed points, and the brain structure-function relationships with the parameter heterogeneity affected by neurodegenerative diseases, remain to be explored in future studies.

Information sharing statement

The patient data sets cannot be made publicly available due to the data protection concerns regarding potentially identifying and sensitive patient information. Interested researchers may access the data sets by contacting Clinical Data Manager Aurélie Ponz

(aurelie.ponz@univ-amu.fr) at the Institut de Neurosciences des Systèmes, Aix-Marseille Université. All code is provided freely and is available at GitHub (<https://github.com/ins-amu/BVEP>) and TVB services on the cloud research platform EBRAINS (wiki.ebrains.eu/bin/view/Collabs/bayesian-virtual-epileptic-patient).

CRediT authorship contribution statement

Meysam Hashemi: Conceptualization, Methodology, Software, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Anirudh N. Vattikonda:** Methodology, Visualization, Writing – review & editing. **Jayant Jha:** Methodology, Writing – review & editing. **Viktor Sip:** Methodology, Writing – review & editing. **Marmaduke M. Woodman:** Methodology, Writing – review & editing. **Fabrice Bartolomei:** Data curation, Funding acquisition, Writing – review & editing. **Viktor K. Jirsa:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.neunet.2023.03.040>.

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